

Complete Course: Differential Calculus

Mathematical Foundations for Data Science and Engineering

Contents

1	Introduction to Calculus	4
1.1	Historical Context	4
1.2	The Two Main Branches	4
1.3	Real Number System	4
2	Limits and Continuity	4
2.1	The Limit Concept	4
2.2	Limit Theorems	4
2.3	Important Limits	5
2.4	Continuity	6
2.5	Important Theorems on Continuity	6
3	The Derivative	6
3.1	Definition and Basic Concepts	6
3.2	Geometric Interpretation	6
3.3	Basic Differentiation Rules	7
3.4	Higher-Order Derivatives	7
3.5	Derivatives of Elementary Functions	8
3.6	Implicit Differentiation	8
3.7	Logarithmic Differentiation	9
4	Applications of Derivatives	9
4.1	Related Rates	9
4.2	Linear Approximation and Differentials	9
4.3	Mean Value Theorem and Consequences	10
4.4	Monotonicity and First Derivative Test	10
4.5	Concavity and Second Derivative Test	10
4.6	Optimization Problems	11
4.7	L'Hôpital's Rule	11
5	Taylor Series and Approximations	12
5.1	Taylor Polynomials	12
5.2	Common Taylor Series	13
5.3	Applications of Taylor Series	13

6	Multivariable Calculus	13
6.1	Functions of Several Variables	13
6.2	Partial Derivatives	13
6.3	Directional Derivatives	14
6.4	Total Differential and Differentiability	14
6.5	Chain Rule for Several Variables	14
6.6	Higher-Order Partial Derivatives	14
6.7	Taylor Series in Several Variables	15
7	Optimization in Several Variables	15
7.1	Critical Points	15
7.2	Hessian Matrix	15
7.3	Second Derivative Test	15
7.4	Lagrange Multipliers	16
8	Vector Calculus	16
8.1	Vector Fields	16
8.2	Divergence and Curl	16
8.3	Line Integrals	17
8.4	Fundamental Theorem for Line Integrals	17
8.5	Green's Theorem	17
9	Differential Equations	17
9.1	First-Order Differential Equations	17
9.1.1	Separable Equations	17
9.1.2	Linear First-Order Equations	17
9.1.3	Exact Equations	18
9.2	Second-Order Linear Equations	18
9.2.1	Homogeneous Case ($r(x) = 0$)	18
9.2.2	Constant Coefficients	18
9.2.3	Nonhomogeneous Case	18
9.3	Systems of Differential Equations	18
10	Applications in Data Science	19
10.1	Gradient Descent	19
10.2	Newton's Method	19
10.3	Backpropagation in Neural Networks	19
11	Numerical Differentiation	20
11.1	Finite Difference Formulas	20
11.2	Richardson Extrapolation	20
12	Advanced Topics	20
12.1	Implicit Function Theorem	20
12.2	Inverse Function Theorem	20
12.3	Contraction Mapping Theorem	20

13 Exercises	21
13.1 Basic Exercises	21
13.2 Intermediate Exercises	21
13.3 Advanced Exercises	21

1 Introduction to Calculus

1.1 Historical Context

Calculus, independently discovered by Isaac Newton and Gottfried Wilhelm Leibniz in the late 17th century, revolutionized mathematics and science. It provides tools for studying change and motion, and has become fundamental to physics, engineering, economics, and data science.

1.2 The Two Main Branches

- **Differential Calculus:** Studies rates of change and slopes of curves
- **Integral Calculus:** Studies accumulation and areas under curves

1.3 Real Number System

Definition 1.1 (Real Numbers). *The set of real numbers $\mathbb{R}$ is a complete ordered field satisfying:*

1. *Field axioms (addition and multiplication properties)*
2. *Order axioms ($<, >, \leq, \geq$ relationships)*
3. *Completeness axiom: Every nonempty subset that is bounded above has a least upper bound*

2 Limits and Continuity

2.1 The Limit Concept

Definition 2.1 (Limit of a Function). *Let $f : D \rightarrow \mathbb{R}$ be a function and let c be a limit point of D . We say that $\lim_{x \rightarrow c} f(x) = L$ if:*

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } 0 < |x - c| < \delta \implies |f(x) - L| < \epsilon$$

Definition 2.2 (One-Sided Limits).

$$\begin{aligned} \lim_{x \rightarrow c^+} f(x) = L &\iff \forall \epsilon > 0, \exists \delta > 0 : 0 < x - c < \delta \implies |f(x) - L| < \epsilon \\ \lim_{x \rightarrow c^-} f(x) = L &\iff \forall \epsilon > 0, \exists \delta > 0 : 0 < c - x < \delta \implies |f(x) - L| < \epsilon \end{aligned}$$

2.2 Limit Theorems

Theorem 2.3 (Limit Laws). *If $\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} g(x) = M$, then:*

1. $\lim_{x \rightarrow c} [f(x) + g(x)] = L + M$

$$2. \lim_{x \rightarrow c} [f(x) - g(x)] = L - M$$

$$3. \lim_{x \rightarrow c} [kf(x)] = kL \text{ for any constant } k$$

$$4. \lim_{x \rightarrow c} [f(x)g(x)] = LM$$

$$5. \lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{L}{M} \text{ if } M \neq 0$$

$$6. \lim_{x \rightarrow c} [f(x)]^n = L^n \text{ for } n \in \mathbb{N}$$

Proof of Sum Rule. Given $\epsilon > 0$, since $\lim_{x \rightarrow c} f(x) = L$, $\exists \delta_1 > 0$ such that $0 < |x - c| < \delta_1 \implies |f(x) - L| < \epsilon/2$. Since $\lim_{x \rightarrow c} g(x) = M$, $\exists \delta_2 > 0$ such that $0 < |x - c| < \delta_2 \implies |g(x) - M| < \epsilon/2$. Take $\delta = \min(\delta_1, \delta_2)$. Then for $0 < |x - c| < \delta$:

$$|(f(x) + g(x)) - (L + M)| \leq |f(x) - L| + |g(x) - M| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

□

2.3 Important Limits

Theorem 2.4 (Special Limits).

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin x}{x} &= 1 \\ \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} &= 0 \\ \lim_{x \rightarrow 0} \frac{e^x - 1}{x} &= 1 \\ \lim_{x \rightarrow 0} \frac{\ln(1 + x)}{x} &= 1 \\ \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x &= e \\ \lim_{x \rightarrow 0} (1 + x)^{1/x} &= e \end{aligned}$$

Proof that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$. Consider the unit circle. For $0 < x < \pi/2$, we have:

$$\sin x < x < \tan x$$

Dividing by $\sin x > 0$:

$$1 < \frac{x}{\sin x} < \frac{1}{\cos x}$$

Taking reciprocals:

$$\cos x < \frac{\sin x}{x} < 1$$

By the Squeeze Theorem and continuity of $\cos x$, we get the result.

□

2.4 Continuity

Definition 2.5 (Continuity at a Point). A function f is **continuous** at c if:

1. $f(c)$ is defined
2. $\lim_{x \rightarrow c} f(x)$ exists
3. $\lim_{x \rightarrow c} f(x) = f(c)$

Definition 2.6 (Types of Continuity). • **Continuous on an interval:** Continuous at every point in the interval

- **Uniformly continuous:** $\forall \epsilon > 0, \exists \delta > 0$ such that for all x, y in domain: $|x - y| < \delta \implies |f(x) - f(y)| < \epsilon$
- **Lipschitz continuous:** $\exists K > 0$ such that $|f(x) - f(y)| \leq K|x - y|$ for all x, y

2.5 Important Theorems on Continuity

Theorem 2.7 (Intermediate Value Theorem). If f is continuous on $[a, b]$ and k is between $f(a)$ and $f(b)$, then $\exists c \in (a, b)$ such that $f(c) = k$.

Theorem 2.8 (Extreme Value Theorem). If f is continuous on a closed interval $[a, b]$, then f attains both a maximum and minimum value on $[a, b]$.

Theorem 2.9 (Bolzano-Weierstrass). Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

3 The Derivative

3.1 Definition and Basic Concepts

Definition 3.1 (Derivative). The **derivative** of f at x is:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

provided this limit exists. Alternative notation: $\frac{df}{dx}$, $Df(x)$, $\dot{f}(x)$.

Definition 3.2 (Differentiability). A function f is **differentiable** at x if $f'(x)$ exists. It is differentiable on an interval if differentiable at every point.

3.2 Geometric Interpretation

- Derivative = slope of tangent line at point $(x, f(x))$
- Tangent line equation: $y = f(a) + f'(a)(x - a)$
- Derivative = instantaneous rate of change

3.3 Basic Differentiation Rules

Theorem 3.3 (Differentiation Rules). 1. **Constant Rule:** $\frac{d}{dx}[c] = 0$

2. **Power Rule:** $\frac{d}{dx}[x^n] = nx^{n-1}$ for $n \in \mathbb{R}$

3. **Constant Multiple:** $\frac{d}{dx}[cf(x)] = cf'(x)$

4. **Sum Rule:** $\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$

5. **Product Rule:** $\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$

6. **Quotient Rule:** $\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}, g(x) \neq 0$

7. **Chain Rule:** $\frac{d}{dx}[f(g(x))] = f'(g(x))g'(x)$

Proof of Product Rule.

$$\begin{aligned} \frac{d}{dx}[f(x)g(x)] &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x+h)g(x) + f(x+h)g(x) - f(x)g(x)}{h} \\ &= \lim_{h \rightarrow 0} \left[f(x+h) \frac{g(x+h) - g(x)}{h} + g(x) \frac{f(x+h) - f(x)}{h} \right] \\ &= f(x)g'(x) + g(x)f'(x) \end{aligned}$$

□

Proof of Chain Rule. Let $y = f(u)$ and $u = g(x)$. Then:

$$\begin{aligned} \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{f(g(x+h)) - f(g(x))}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(g(x+h)) - f(g(x))}{g(x+h) - g(x)} \cdot \frac{g(x+h) - g(x)}{h} \\ &= f'(g(x))g'(x) \end{aligned}$$

provided $g(x+h) - g(x) \neq 0$ for small $h \neq 0$. A more rigorous proof handles the case when $g(x+h) = g(x)$. □

3.4 Higher-Order Derivatives

Definition 3.4 (Higher Derivatives).

$$\begin{aligned} f''(x) &= \frac{d}{dx}[f'(x)] = \frac{d^2 f}{dx^2} \\ f^{(n)}(x) &= \frac{d}{dx}[f^{(n-1)}(x)] = \frac{d^n f}{dx^n} \end{aligned}$$

3.5 Derivatives of Elementary Functions

Theorem 3.5 (Basic Derivatives).

$$\begin{aligned}\frac{d}{dx}[e^x] &= e^x \\ \frac{d}{dx}[a^x] &= a^x \ln a \quad (a > 0) \\ \frac{d}{dx}[\ln x] &= \frac{1}{x} \quad (x > 0) \\ \frac{d}{dx}[\log_a x] &= \frac{1}{x \ln a} \quad (a > 0, a \neq 1, x > 0) \\ \frac{d}{dx}[\sin x] &= \cos x \\ \frac{d}{dx}[\cos x] &= -\sin x \\ \frac{d}{dx}[\tan x] &= \sec^2 x \\ \frac{d}{dx}[\cot x] &= -\csc^2 x \\ \frac{d}{dx}[\sec x] &= \sec x \tan x \\ \frac{d}{dx}[\csc x] &= -\csc x \cot x\end{aligned}$$

Proof that $\frac{d}{dx}[e^x] = e^x$

$$\begin{aligned}\frac{d}{dx}[e^x] &= \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h} \\ &= e^x \lim_{h \rightarrow 0} \frac{e^h - 1}{h} \\ &= e^x \cdot 1 = e^x\end{aligned}$$

using $\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$. □

3.6 Implicit Differentiation

Theorem 3.6 (Implicit Differentiation). *Given an equation $F(x, y) = 0$, we can find $\frac{dy}{dx}$ by differentiating both sides with respect to x and solving for $\frac{dy}{dx}$.*

Example 3.7. For $x^2 + y^2 = 1$:

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(1) \implies 2x + 2y \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{x}{y}$$

3.7 Logarithmic Differentiation

[Logarithmic Differentiation] For functions of the form $y = [f(x)]^{g(x)}$ or products/quotients with many factors:

1. Take natural log of both sides: $\ln y = \ln(f(x))$ or $\ln y = g(x) \ln(f(x))$
2. Differentiate implicitly
3. Solve for y'

4 Applications of Derivatives

4.1 Related Rates

[Related Rates Problems]

1. Identify given rates and rate to find
2. Write equation relating variables
3. Differentiate with respect to time
4. Substitute known values and solve

Example 4.1 (Filling a Sphere). *Water flows into a spherical tank at rate k . The volume $V = \frac{4}{3}\pi r^3$. Find $\frac{dr}{dt}$ when $r = R$:*

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} \implies \frac{dr}{dt} = \frac{k}{4\pi R^2}$$

4.2 Linear Approximation and Differentials

Definition 4.2 (Linear Approximation). *The linear approximation of f near $x = a$ is:*

$$L(x) = f(a) + f'(a)(x - a)$$

Definition 4.3 (Differential). *For $y = f(x)$, the differential $dy = f'(x)dx$, where dx is an increment in x .*

Theorem 4.4 (Error in Linear Approximation). *The error $E(x) = f(x) - L(x)$ satisfies:*

$$\lim_{x \rightarrow a} \frac{E(x)}{x - a} = 0$$

4.3 Mean Value Theorem and Consequences

Theorem 4.5 (Rolle's Theorem). *If f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then $\exists c \in (a, b)$ such that $f'(c) = 0$.*

Theorem 4.6 (Mean Value Theorem (MVT)). *If f is continuous on $[a, b]$ and differentiable on (a, b) , then $\exists c \in (a, b)$ such that:*

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Proof. Let $g(x) = f(x) - \left[f(a) + \frac{f(b)-f(a)}{b-a}(x-a) \right]$. Then $g(a) = g(b) = 0$. By Rolle's Theorem, $\exists c \in (a, b)$ such that $g'(c) = 0$, which gives the result. $\square$

Theorem 4.7 (Cauchy's Mean Value Theorem). *If f and g are continuous on $[a, b]$ and differentiable on (a, b) , and $g'(x) \neq 0$ on (a, b) , then $\exists c \in (a, b)$ such that:*

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

4.4 Monotonicity and First Derivative Test

Theorem 4.8 (Monotonicity). *Let f be continuous on $[a, b]$ and differentiable on (a, b) .*

- 1. If $f'(x) > 0$ on (a, b) , then f is strictly increasing on $[a, b]$*
- 2. If $f'(x) < 0$ on (a, b) , then f is strictly decreasing on $[a, b]$*

Theorem 4.9 (First Derivative Test). *Let c be a critical point of f (where $f'(c) = 0$ or $f'(c)$ doesn't exist).*

- 1. If f' changes from positive to negative at c , then f has a local maximum at c*
- 2. If f' changes from negative to positive at c , then f has a local minimum at c*
- 3. If f' doesn't change sign at c , then f has neither maximum nor minimum at c*

4.5 Concavity and Second Derivative Test

Definition 4.10 (Concavity). *A function f is:*

- **Concave up** on an interval if $f''(x) > 0$ (holds water)*
- **Concave down** on an interval if $f''(x) < 0$ (spills water)*

Definition 4.11 (Inflection Point). *A point where f changes concavity.*

Theorem 4.12 (Second Derivative Test). *Let $f'(c) = 0$.*

- 1. If $f''(c) > 0$, then f has a local minimum at c*
- 2. If $f''(c) < 0$, then f has a local maximum at c*
- 3. If $f''(c) = 0$, the test is inconclusive*

4.6 Optimization Problems

[Solving Optimization Problems]

1. Understand the problem and identify quantities
2. Draw diagram if applicable
3. Introduce variables and write objective function
4. Identify constraints
5. Express objective function in one variable
6. Find domain
7. Find critical points
8. Use First/Second Derivative Test or check endpoints
9. Interpret solution

Example 4.13 (Box with Maximum Volume). *From a 12×12 inch square, cut corners of size x to make open box. Volume:*

$$V(x) = x(12 - 2x)^2, \quad 0 < x < 6$$

$$V'(x) = (12 - 2x)^2 - 4x(12 - 2x) = 4(6 - x)(6 - 3x)$$

Critical points: $x = 2$ (maximum since $V''(2) < 0$), $x = 6$ (minimum).

4.7 L'Hôpital's Rule

Theorem 4.14 (L'Hôpital's Rule). *If $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$ or $\pm\infty$, and $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ exists, then:*

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Sketch for 0/0 case. By Cauchy's MVT, for x near a :

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f'(c)}{g'(c)}$$

for some c between a and x . As $x \rightarrow a$, $c \rightarrow a$. □

5 Taylor Series and Approximations

5.1 Taylor Polynomials

Definition 5.1 (Taylor Polynomial). *The n th degree Taylor polynomial for f at a is:*

$$T_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x - a)^k$$

Theorem 5.2 (Taylor's Theorem with Remainder). *If f has $n + 1$ continuous derivatives on interval containing a , then for x in this interval:*

$$f(x) = T_n(x) + R_n(x)$$

where the remainder $R_n(x)$ can be written as:

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} (x - a)^{n+1} \quad (\text{Lagrange form})$$

for some c between a and x .

Using Integration by Parts.

$$\begin{aligned} f(x) &= f(a) + \int_a^x f'(t) dt \\ &= f(a) + [-(x-t)f'(t)]_a^x + \int_a^x (x-t)f''(t) dt \\ &= f(a) + f'(a)(x-a) + \int_a^x (x-t)f''(t) dt \\ &\dots \\ &= T_n(x) + \int_a^x \frac{(x-t)^n}{n!} f^{(n+1)}(t) dt \end{aligned}$$

Then apply Mean Value Theorem for Integrals. □

5.2 Common Taylor Series

Theorem 5.3 (Important Taylor Series).

$$\begin{aligned}e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \quad (-\infty < x < \infty) \\ \sin x &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \quad (-\infty < x < \infty) \\ \cos x &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots \quad (-\infty < x < \infty) \\ \ln(1+x) &= \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots \quad (-1 < x \leq 1) \\ (1+x)^\alpha &= \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \cdots \quad (-1 < x < 1)\end{aligned}$$

5.3 Applications of Taylor Series

1. **Approximating functions:** Polynomials are easier to compute
2. **Solving differential equations:** Power series method
3. **Evaluating limits:** More powerful than L'Hôpital's for some cases
4. **Error analysis:** Bounding approximation errors

6 Multivariable Calculus

6.1 Functions of Several Variables

Definition 6.1 (Function of Several Variables). *A function $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}$ assigns a real number to each point $(x_1, \dots, x_n) \in D$.*

Definition 6.2 (Level Sets). *For constant c , the level set is $\{(x_1, \dots, x_n) : f(x_1, \dots, x_n) = c\}$.*

6.2 Partial Derivatives

Definition 6.3 (Partial Derivative). *The partial derivative of f with respect to x_i at $\mathbf{a}$ is:*

$$\frac{\partial f}{\partial x_i}(\mathbf{a}) = \lim_{h \rightarrow 0} \frac{f(a_1, \dots, a_i + h, \dots, a_n) - f(\mathbf{a})}{h}$$

Definition 6.4 (Gradient). *The gradient of f is:*

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right)$$

6.3 Directional Derivatives

Definition 6.5 (Directional Derivative). *The directional derivative of f at $\mathbf{a}$ in direction $\mathbf{u}$ (unit vector) is:*

$$D_{\mathbf{u}}f(\mathbf{a}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{a} + h\mathbf{u}) - f(\mathbf{a})}{h}$$

Theorem 6.6 (Directional Derivative Formula). *If f is differentiable at $\mathbf{a}$, then:*

$$D_{\mathbf{u}}f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{u}$$

6.4 Total Differential and Differentiability

Definition 6.7 (Differentiability in $\mathbb{R}^n$). *$f : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable at $\mathbf{a}$ if there exists a linear transformation L such that:*

$$\lim_{\mathbf{h} \rightarrow 0} \frac{f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) - L(\mathbf{h})}{\|\mathbf{h}\|} = 0$$

Then L is the total differential, $df = \nabla f \cdot d\mathbf{x}$.

Theorem 6.8 (Sufficient Condition for Differentiability). *If all partial derivatives exist and are continuous near $\mathbf{a}$, then f is differentiable at $\mathbf{a}$.*

6.5 Chain Rule for Several Variables

Theorem 6.9 (Chain Rule). *If $z = f(x, y)$ where $x = x(t)$ and $y = y(t)$ are differentiable, then:*

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

Theorem 6.10 (General Chain Rule). *For $z = f(x_1, \dots, x_n)$ with $x_i = x_i(t_1, \dots, t_m)$:*

$$\frac{\partial z}{\partial t_j} = \sum_{i=1}^n \frac{\partial f}{\partial x_i} \frac{\partial x_i}{\partial t_j}$$

6.6 Higher-Order Partial Derivatives

Definition 6.11 (Second Partial Derivatives).

$$\begin{aligned} \frac{\partial^2 f}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) \\ \frac{\partial^2 f}{\partial y \partial x} &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) \end{aligned}$$

Theorem 6.12 (Clairaut's Theorem/Equality of Mixed Partial). *If f_{xy} and f_{yx} are continuous, then $f_{xy} = f_{yx}$.*

6.7 Taylor Series in Several Variables

Theorem 6.13 (Multivariable Taylor Series). *For $f : \mathbb{R}^n \rightarrow \mathbb{R}$ with continuous partial derivatives up to order $k + 1$:*

$$f(\mathbf{a} + \mathbf{h}) = f(\mathbf{a}) + \sum_{|\alpha|=1}^k \frac{D^\alpha f(\mathbf{a})}{\alpha!} \mathbf{h}^\alpha + R_k(\mathbf{h})$$

where $\alpha = (\alpha_1, \dots, \alpha_n)$ multi-index, $|\alpha| = \sum \alpha_i$, $\alpha! = \alpha_1! \cdots \alpha_n!$, $\mathbf{h}^\alpha = h_1^{\alpha_1} \cdots h_n^{\alpha_n}$, and $D^\alpha f = \frac{\partial^{|\alpha|} f}{\partial x_1^{\alpha_1} \cdots \partial x_n^{\alpha_n}}$.

7 Optimization in Several Variables

7.1 Critical Points

Definition 7.1 (Critical Point). *$\mathbf{a}$ is a critical point of f if $\nabla f(\mathbf{a}) = \mathbf{0}$ or $\nabla f(\mathbf{a})$ doesn't exist.*

7.2 Hessian Matrix

Definition 7.2 (Hessian Matrix). *The Hessian of $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is:*

$$H_f(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

7.3 Second Derivative Test

Theorem 7.3 (Second Derivative Test for $\mathbb{R}^2$). *Let f have continuous second partials at (a, b) with $f_x(a, b) = f_y(a, b) = 0$. Define:*

$$D = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2$$

Then:

1. *If $D > 0$ and $f_{xx}(a, b) > 0$: local minimum*
2. *If $D > 0$ and $f_{xx}(a, b) < 0$: local maximum*
3. *If $D < 0$: saddle point*
4. *If $D = 0$: test inconclusive*

Theorem 7.4 (General Second Derivative Test). *For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, at critical point $\mathbf{a}$:*

1. *If $H_f(\mathbf{a})$ is positive definite: local minimum*

2. If $H_f(\mathbf{a})$ is negative definite: local maximum
3. If $H_f(\mathbf{a})$ has both positive and negative eigenvalues: saddle point
4. If $H_f(\mathbf{a})$ is indefinite but has zero eigenvalues: test inconclusive

7.4 Lagrange Multipliers

[Lagrange Multipliers for One Constraint] To optimize $f(\mathbf{x})$ subject to $g(\mathbf{x}) = 0$, solve:

$$\nabla f(\mathbf{x}) = \lambda \nabla g(\mathbf{x}), \quad g(\mathbf{x}) = 0$$

Theorem 7.5 (Lagrange Multipliers Theorem). *If f has an extremum at $\mathbf{a}$ subject to $g(\mathbf{x}) = 0$, and $\nabla g(\mathbf{a}) \neq \mathbf{0}$, then $\exists \lambda$ such that:*

$$\nabla f(\mathbf{a}) = \lambda \nabla g(\mathbf{a})$$

Sketch. The constraint $g(\mathbf{x}) = 0$ defines a surface. At extremum, ∇f must be perpendicular to the tangent plane of this surface, hence parallel to ∇g . $\square$

[Lagrange Multipliers for Multiple Constraints] To optimize $f(\mathbf{x})$ subject to $g_1(\mathbf{x}) = 0, \dots, g_k(\mathbf{x}) = 0$, solve:

$$\nabla f(\mathbf{x}) = \sum_{i=1}^k \lambda_i \nabla g_i(\mathbf{x}), \quad g_i(\mathbf{x}) = 0, \quad i = 1, \dots, k$$

8 Vector Calculus

8.1 Vector Fields

Definition 8.1 (Vector Field). *A vector field on $\mathbb{R}^n$ is a function $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^n$.*

8.2 Divergence and Curl

Definition 8.2 (Divergence). *For $\mathbf{F} = (F_1, F_2, F_3)$ in $\mathbb{R}^3$:*

$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

Definition 8.3 (Curl). *For $\mathbf{F} = (F_1, F_2, F_3)$ in $\mathbb{R}^3$:*

$$\operatorname{curl} \mathbf{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$

8.3 Line Integrals

Definition 8.4 (Line Integral of Scalar Field). For $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and curve C parameterized by $\mathbf{r}(t)$, $a \leq t \leq b$:

$$\int_C f \, ds = \int_a^b f(\mathbf{r}(t)) \|\mathbf{r}'(t)\| dt$$

Definition 8.5 (Line Integral of Vector Field). For $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and curve C :

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$$

8.4 Fundamental Theorem for Line Integrals

Theorem 8.6 (Fundamental Theorem for Line Integrals). If $\mathbf{F} = \nabla f$ is conservative, then for any curve C from A to B :

$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(B) - f(A)$$

8.5 Green's Theorem

Theorem 8.7 (Green's Theorem). For simple closed curve C enclosing region D , and $\mathbf{F} = (P, Q)$:

$$\oint_C P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

9 Differential Equations

9.1 First-Order Differential Equations

Definition 9.1 (First-Order ODE). An equation of the form $F(x, y, y') = 0$ or $y' = f(x, y)$.

9.1.1 Separable Equations

[Separable Equations] If $y' = f(x)g(y)$, then:

$$\int \frac{dy}{g(y)} = \int f(x) dx$$

9.1.2 Linear First-Order Equations

[Linear First-Order ODE] For $y' + P(x)y = Q(x)$, use integrating factor $\mu(x) = e^{\int P(x) dx}$:

$$\frac{d}{dx}[\mu(x)y] = \mu(x)Q(x)$$

9.1.3 Exact Equations

Definition 9.2 (Exact Equation). $M(x, y)dx + N(x, y)dy = 0$ is exact if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

[Solving Exact Equations] Find $F(x, y)$ such that:

$$\frac{\partial F}{\partial x} = M, \quad \frac{\partial F}{\partial y} = N$$

Then solution is $F(x, y) = C$.

9.2 Second-Order Linear Equations

Definition 9.3 (Second-Order Linear ODE). Equation of the form $y'' + p(x)y' + q(x)y = r(x)$.

9.2.1 Homogeneous Case ($r(x) = 0$)

Theorem 9.4 (Superposition Principle). If y_1 and y_2 are solutions, then $c_1y_1 + c_2y_2$ is also a solution.

Theorem 9.5 (General Solution). If y_1 and y_2 are linearly independent solutions, then general solution is $y = c_1y_1 + c_2y_2$.

9.2.2 Constant Coefficients

For $ay'' + by' + cy = 0$, try $y = e^{rx}$:

$$ar^2 + br + c = 0 \quad (\text{characteristic equation})$$

- Distinct real roots $r_1 \neq r_2$: $y = c_1e^{r_1x} + c_2e^{r_2x}$
- Repeated root r : $y = c_1e^{rx} + c_2xe^{rx}$
- Complex roots $\alpha \pm i\beta$: $y = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$

9.2.3 Nonhomogeneous Case

[Method of Undetermined Coefficients] Guess particular solution based on form of $r(x)$.

[Variation of Parameters] If y_1, y_2 are solutions of homogeneous equation, try $y_p = u_1y_1 + u_2y_2$ where:

$$u_1' = -\frac{y_2r}{W}, \quad u_2' = \frac{y_1r}{W}$$

with Wronskian $W = y_1y_2' - y_2y_1'$.

9.3 Systems of Differential Equations

[Matrix Method] For $\mathbf{x}' = A\mathbf{x}$, solution is $\mathbf{x} = e^{At}\mathbf{x}_0$.

10 Applications in Data Science

10.1 Gradient Descent

Algorithm 1 Gradient Descent

Require: Objective function f , initial point $\mathbf{x}_0$, learning rate η , tolerance ϵ

Ensure: Approximate minimum $\mathbf{x}^*$

- 1: Initialize $\mathbf{x} \leftarrow \mathbf{x}_0$
 - 2: **repeat**
 - 3: Compute gradient: $\mathbf{g} \leftarrow \nabla f(\mathbf{x})$
 - 4: Update: $\mathbf{x} \leftarrow \mathbf{x} - \eta \mathbf{g}$
 - 5: **until** $\|\mathbf{g}\| < \epsilon$ or maximum iterations reached
 - 6: **return** $\mathbf{x}$
-

Theorem 10.1 (Convergence of Gradient Descent). *For convex f with L -Lipschitz gradient, gradient descent with $\eta = 1/L$ converges to global minimum.*

10.2 Newton's Method

Algorithm 2 Newton's Method for Optimization

Require: Twice-differentiable f , initial point $\mathbf{x}_0$, tolerance ϵ

Ensure: Approximate critical point $\mathbf{x}^*$

- 1: Initialize $\mathbf{x} \leftarrow \mathbf{x}_0$
 - 2: **repeat**
 - 3: Compute gradient: $\mathbf{g} \leftarrow \nabla f(\mathbf{x})$
 - 4: Compute Hessian: $H \leftarrow H_f(\mathbf{x})$
 - 5: Solve $H\mathbf{d} = -\mathbf{g}$ for $\mathbf{d}$
 - 6: Update: $\mathbf{x} \leftarrow \mathbf{x} + \mathbf{d}$
 - 7: **until** $\|\mathbf{g}\| < \epsilon$ or maximum iterations reached
 - 8: **return** $\mathbf{x}$
-

10.3 Backpropagation in Neural Networks

Theorem 10.2 (Chain Rule for Backpropagation). *For neural network with loss L , weights W , and activations a :*

$$\frac{\partial L}{\partial W_{ij}} = \frac{\partial L}{\partial a_{next}} \frac{\partial a_{next}}{\partial W_{ij}}$$

Backpropagation computes gradients efficiently using dynamic programming.

11 Numerical Differentiation

11.1 Finite Difference Formulas

Theorem 11.1 (Finite Difference Approximations).

$$\begin{aligned}f'(x) &\approx \frac{f(x+h) - f(x)}{h} && (\text{Forward difference, error } O(h)) \\f'(x) &\approx \frac{f(x) - f(x-h)}{h} && (\text{Backward difference, error } O(h)) \\f'(x) &\approx \frac{f(x+h) - f(x-h)}{2h} && (\text{Central difference, error } O(h^2)) \\f''(x) &\approx \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} && (\text{Error } O(h^2))\end{aligned}$$

11.2 Richardson Extrapolation

[Richardson Extrapolation] To improve accuracy:

$$D_h = \frac{4D_{h/2} - D_h}{3} \quad \text{for central difference (error reduces from } O(h^2) \text{ to } O(h^4))$$

12 Advanced Topics

12.1 Implicit Function Theorem

Theorem 12.1 (Implicit Function Theorem). *If $F : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^m$ is C^1 , $F(\mathbf{a}, \mathbf{b}) = \mathbf{0}$, and $\det \left(\frac{\partial F_i}{\partial y_j}(\mathbf{a}, \mathbf{b}) \right) \neq 0$, then near $(\mathbf{a}, \mathbf{b})$, equation $F(\mathbf{x}, \mathbf{y}) = \mathbf{0}$ defines $\mathbf{y} = f(\mathbf{x})$ with f differentiable and:*

$$\frac{\partial f}{\partial x_i} = - \left[\frac{\partial F}{\partial \mathbf{y}} \right]^{-1} \frac{\partial F}{\partial x_i}$$

12.2 Inverse Function Theorem

Theorem 12.2 (Inverse Function Theorem). *If $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is C^1 and $\det(J_f(\mathbf{a})) \neq 0$, then f has a local inverse f^{-1} near $\mathbf{a}$ with:*

$$J_{f^{-1}}(f(\mathbf{a})) = [J_f(\mathbf{a})]^{-1}$$

12.3 Contraction Mapping Theorem

Theorem 12.3 (Contraction Mapping/Banach Fixed Point). *If $T : X \rightarrow X$ is a contraction on complete metric space X (i.e., $d(T(x), T(y)) \leq kd(x, y)$ with $0 \leq k < 1$), then T has a unique fixed point.*

13 Exercises

13.1 Basic Exercises

Exercise 13.1. Find derivative of $f(x) = x^3 \sin(2x)$ using product rule.

Exercise 13.2. Find $\frac{dy}{dx}$ for $x^2 + xy + y^2 = 3$ using implicit differentiation.

Exercise 13.3. Find critical points of $f(x) = x^3 - 3x^2 + 4$ and classify them.

13.2 Intermediate Exercises

Exercise 13.4. Find Taylor polynomial of degree 3 for $f(x) = \ln(1+x)$ at $x = 0$.

Exercise 13.5. Find and classify critical points of $f(x, y) = x^3 + y^3 - 3xy$.

Exercise 13.6. Use Lagrange multipliers to find extreme values of $f(x, y) = x^2 + y^2$ on line $x + y = 1$.

13.3 Advanced Exercises

Exercise 13.7. Prove that if f is differentiable and $f'(x) > 0$ for all x , then f is strictly increasing.

Exercise 13.8. Use implicit function theorem to show y can be expressed as function of x near $(1, 1)$ for $x^3 + y^3 = 2xy$.

Exercise 13.9. Implement gradient descent to minimize $f(x) = x^2 + \sin(x)$.

Appendix: Useful Formulas and Tables

Derivative Rules Summary

Function	Derivative
c	0
x^n	nx^{n-1}
e^x	e^x
a^x	$a^x \ln a$
$\ln x$	$\frac{1}{x}$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\csc^2 x$
$\sec x$	$\sec x \tan x$
$\csc x$	$-\csc x \cot x$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$
$\cos^{-1} x$	$-\frac{1}{\sqrt{1-x^2}}$
$\tan^{-1} x$	$\frac{1}{1+x^2}$

Table 1: Basic Derivatives

Integration Formulas

Function	Integral
x^n	$\frac{x^{n+1}}{n+1} + C, n \neq -1$
$\frac{1}{x}$	$\ln x + C$
e^x	$e^x + C$
$\sin x$	$-\cos x + C$
$\cos x$	$\sin x + C$
$\sec^2 x$	$\tan x + C$
$\frac{1}{1+x^2}$	$\tan^{-1} x + C$
$\frac{1}{\sqrt{1-x^2}}$	$\sin^{-1} x + C$

Table 2: Basic Integrals

Taylor Series Expansions

Function	Expansion	Interval
e^x	$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$	$(-\infty, \infty)$
$\sin x$	$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$	$(-\infty, \infty)$
$\cos x$	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$	$(-\infty, \infty)$
$\ln(1+x)$	$x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$	$(-1, 1]$
$(1+x)^\alpha$	$1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \dots$	$(-1, 1)$
$\frac{1}{1-x}$	$1 + x + x^2 + x^3 + \dots$	$(-1, 1)$

Table 3: Common Taylor Series

Differential Equations Solutions

Equation	General Solution
$y' + p(x)y = q(x)$	$y = e^{-\int p dx} \left[\int e^{\int p dx} q dx + C \right]$
$y'' + ay' + by = 0$	See characteristic equation cases
$y' = ky$	$y = Ce^{kx}$
$y' = f(x)g(y)$	$\int \frac{dy}{g(y)} = \int f(x)dx + C$

Table 4: Differential Equations Solutions

Vector Calculus Identities

$$\begin{aligned}
 \nabla(fg) &= f\nabla g + g\nabla f \\
 \nabla \cdot (f\mathbf{F}) &= f\nabla \cdot \mathbf{F} + \mathbf{F} \cdot \nabla f \\
 \nabla \times (f\mathbf{F}) &= f\nabla \times \mathbf{F} + \nabla f \times \mathbf{F} \\
 \nabla \cdot (\mathbf{F} \times \mathbf{G}) &= \mathbf{G} \cdot (\nabla \times \mathbf{F}) - \mathbf{F} \cdot (\nabla \times \mathbf{G}) \\
 \nabla \times (\nabla \times \mathbf{F}) &= \nabla(\nabla \cdot \mathbf{F}) - \nabla^2 \mathbf{F} \\
 \nabla \cdot (\nabla \times \mathbf{F}) &= 0 \\
 \nabla \times (\nabla f) &= 0
 \end{aligned}$$

Useful Inequalities

$$\begin{aligned}
 |\mathbf{a} \cdot \mathbf{b}| &\leq \|\mathbf{a}\| \|\mathbf{b}\| \quad (\text{Cauchy-Schwarz}) \\
 \|\mathbf{a} + \mathbf{b}\| &\leq \|\mathbf{a}\| + \|\mathbf{b}\| \quad (\text{Triangle}) \\
 |f(b) - f(a)| &\leq \sup_{[a,b]} |f'(x)| \cdot |b - a| \quad (\text{Mean Value Inequality}) \\
 \left| \int_a^b f(x)dx \right| &\leq \int_a^b |f(x)|dx
 \end{aligned}$$

Common Limits

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{\ln(1 + x)}{x} = 1$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$$

$$\lim_{x \rightarrow 0} (1 + x)^{1/x} = e$$

References